

MTUNZINI, South Africa

WMO: 684970

Lat: 28.950S

Lon: 31.700E

Elev: 38

StdP: 100.87

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	8.3	9.6	4.5	5.2	17.5	6.2	5.9	15.1	9.2	16.0	8.2	19.6	0.9	250	0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.9	35.0	24.5	33.2	23.9	31.6	23.4	26.4	31.1	25.6	30.1	25.0	29.3	5.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.5	28.2	24.5	19.5	27.5	23.8	18.8	26.9	82.4	31.3	79.0	30.5	76.4	29.4	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.2	7.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			5.3	27.8	1.6	0.9	4.2	28.4	3.2	28.9	2.3	29.4	1.1	30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.9	25.2	25.3	24.6	22.5	20.2	17.9	18.0	19.7	20.7	21.5	22.6	24.4		
	DBStd	3.82	2.79	2.38	2.68	2.78	2.41	2.32	2.48	3.20	3.32	3.20	3.36	3.05		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	128	0	0	1	2	8	34	36	21	13	7	4	1		
	CDD10.0	4337	471	428	454	377	317	238	249	301	320	357	379	447		
	CDD18.3	1424	213	194	196	129	67	22	27	63	83	106	133	190		
	CDH23.3	9289	1678	1429	1326	661	298	115	133	354	463	578	854	1400		
(13)	CDH26.7	2889	558	427	397	184	62	17	22	111	166	193	292	460		
(14)	Wind	WSAvg	3.1	3.4	3.0	2.9	2.7	2.2	2.5	3.1	3.5	3.9	3.9	3.6		
(15) Precipitation	PrecAvg	941	122	115	103	66	40	32	48	36	64	106	110	99		
	PrecMax	1399	240	263	206	212	115	172	408	113	252	233	313	201		
	PrecMin	563	21	22	22	3	0	1	0	3	7	23	16	32		
	PrecStd	211	59	55	49	45	31	37	79	29	47	51	62	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.6	36.0	35.4	33.6	30.8	28.7	29.2	32.7	34.3	34.8	35.8	36.8		
		MCWB	25.2	25.3	24.9	23.3	20.7	17.6	17.0	18.8	20.5	22.8	23.7	25.8		
	2%	DB	33.8	33.1	32.6	30.8	28.3	26.1	26.6	29.6	30.9	31.3	32.6	33.4		
		MCWB	25.0	24.8	24.4	22.8	20.1	17.6	16.9	18.4	20.1	22.0	23.6	25.4		
	5%	DB	31.4	30.9	30.5	28.7	26.3	24.3	24.5	27.1	28.2	28.6	30.0	30.9		
		MCWB	24.7	24.3	24.1	22.3	19.9	17.4	16.8	17.9	19.6	21.3	22.9	24.2		
	10%	DB	29.5	29.1	28.7	26.9	24.7	22.5	22.6	24.7	25.7	26.3	27.7	28.8		
		MCWB	24.2	23.9	23.7	21.9	19.4	17.1	16.6	17.6	19.4	20.7	22.4	23.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	26.7	27.1	25.2	22.5	20.4	19.4	21.1	22.7	24.2	25.5	27.8		
		MCDB	31.4	32.1	30.7	29.5	27.1	24.3	23.8	28.1	29.0	30.4	32.5	32.7		
	2%	WB	26.0	25.8	25.7	24.0	21.3	19.3	18.6	20.0	21.5	22.9	24.3	26.2		
		MCDB	30.6	30.5	29.5	28.1	25.9	23.2	23.1	25.6	27.1	28.9	29.9	31.2		
	5%	WB	25.2	25.0	25.0	23.2	20.6	18.5	17.9	19.2	20.6	22.1	23.6	25.1		
		MCDB	29.6	29.2	28.8	27.1	24.8	22.3	22.3	24.1	25.9	26.8	28.3	29.4		
	10%	WB	24.6	24.5	24.2	22.5	19.9	17.7	17.2	18.5	19.9	21.3	22.8	24.3		
		MCDB	28.7	28.1	27.5	25.9	23.8	21.3	21.4	22.9	24.3	25.3	26.8	28.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	7.9	8.1	8.7	10.4	11.6	11.4	10.3	8.9	7.9	7.9	7.8		
		MCDBR	11.7	11.1	11.3	11.7	12.7	13.7	13.9	14.0	13.8	13.0	12.8	12.2		
	5% WB	MCWBR	4.4	4.0	4.3	4.5	5.7	6.4	5.7	4.9	4.8	5.0	5.1	5.0		
		MCWBR	9.7	9.2	9.8	9.8	11.1	11.7	11.7	11.2	11.0	10.7	10.4	10.4		
(40) Clear-Sky Solar Irradiance	taub		0.447	0.431	0.430	0.405	0.377	0.363	0.370	0.433	0.484	0.500	0.469	0.458		
		taud	2.323	2.370	2.360	2.388	2.428	2.444	2.414	2.227	2.109	2.107	2.229	2.275		
	Ebn at Noon		899	895	863	833	809	799	807	790	798	824	873	891		
		Edh at Noon	137	127	122	108	93	87	93	124	154	164	150	145		
(44) All-Sky Solar Radiation	RadAvg		6.03	5.91	5.15	4.15	3.57	3.17	3.37	3.92	4.49	4.89	5.50	6.00		
(45)	RadStd		0.52	0.45	0.29	0.24	0.17	0.15	0.21	0.25	0.28	0.40	0.50	0.43		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page